GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning

Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziems, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Alexandros G. Dimakis, Ion Stoica, Dan Klein, Matei Zaharia, Omar Khattab

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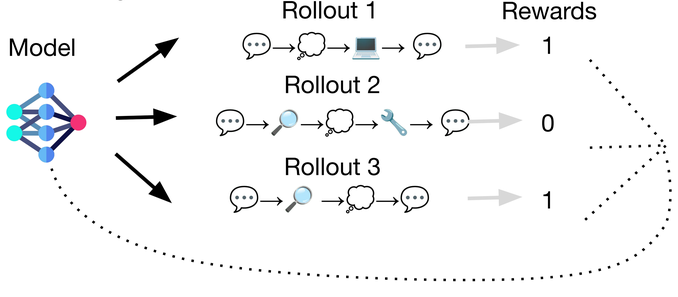

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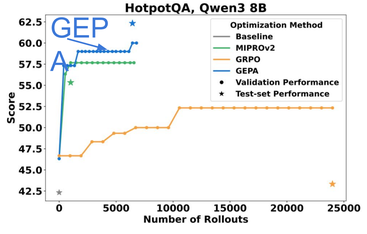
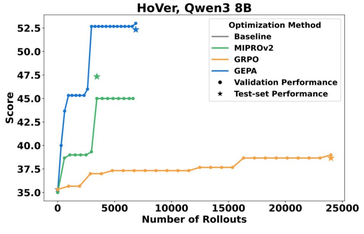


(a) HotpotQA, Qwen3 8B



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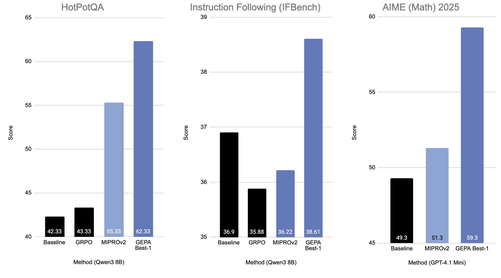

Qwen3 8B	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23	—
GRPO	43.33	35.88	38.67	86.66	**38.00**	51.26	48.91	+3.68
MIPROv2	55.33	36.22	47.33	81.55	20.00	46.60	47.84	+2.61
GEPA	62.33	**38.61**	**52.33**	**91.85**	32.00	**51.95**	**54.85**	**+9.62**
GEPA+Merge	**64.33**	28.23	51.67	86.26	32.00	**51.95**	52.40	+7.17
Total optimization budget (# rollouts)								
GEPA (+Merge)	6871	3593	7051	2426	1839	1839	3936	—
GRPO	24000	24000	24000	24000	24000	24000	24000	—

GPT-4.1 Mini	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	38.00	47.79	46.33	78.57	49.33	58.20	53.03	—
Trace (OptoPrime)	60.33	51.19	46.00	74.18	45.33	60.74	56.30	+3.27
MIPROv2-No-Demos	38.00	52.04	51.33	91.85	48.67	60.97	57.14	+4.11
MIPROv2	58.00	49.15	48.33	83.37	51.33	61.84	58.67	+5.64
TextGrad	62.33	48.64	47.67	85.68	46.67	63.84	59.14	+6.11
GEPA	**69.00**	52.72	51.67	94.47	**59.33**	**64.13**	65.22	+12.19
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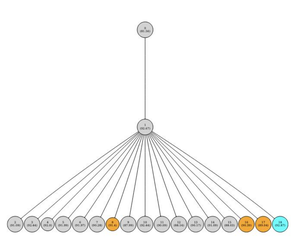
Optimized with Qwen3-8B, evaluated on GPT-4.1-Mini								
GEPA-Qwen-Opt	65.67	49.83	54.67	90.05	52.67	59.31	62.03	+9.00

Pareto-based Candidate Selection is Important



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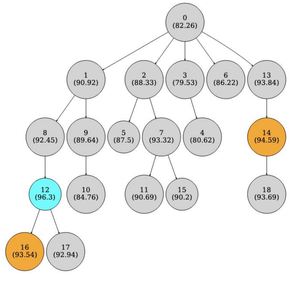

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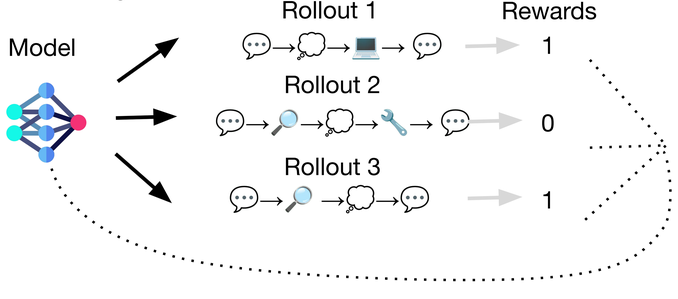

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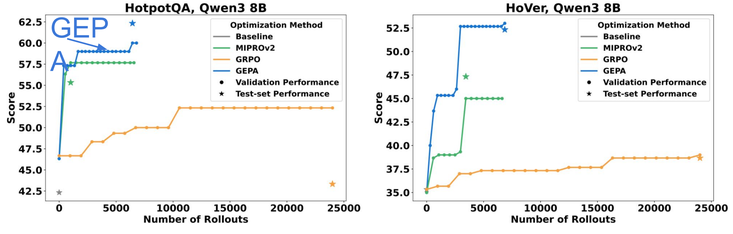

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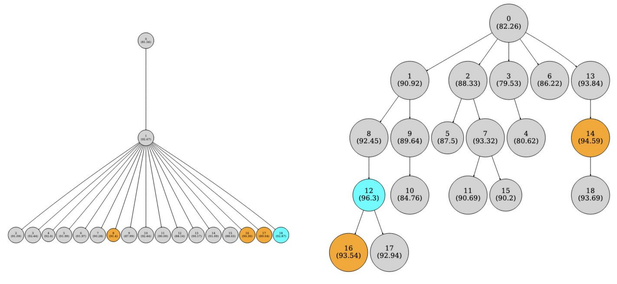

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Prompt optimization to text optimization

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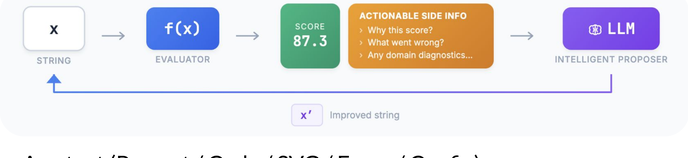

- Any text (Prompt / Code / SVG / Essay / Config)
- Any task (Single-task search, Multi-task search, Generalization)
- Any algorithm (ShinkaEvolve, OpenEvolve, Combee)

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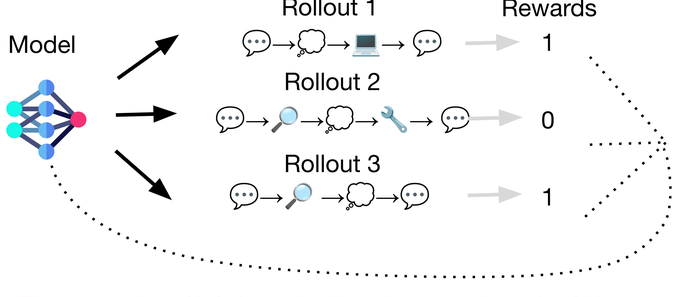

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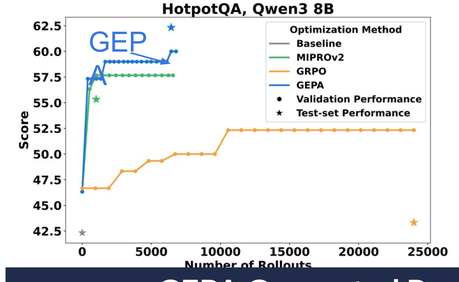

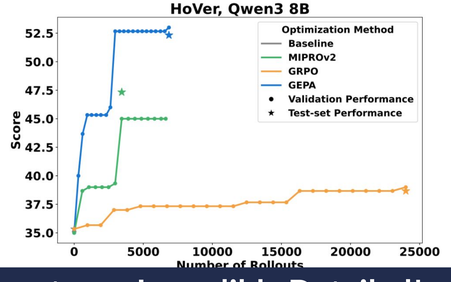

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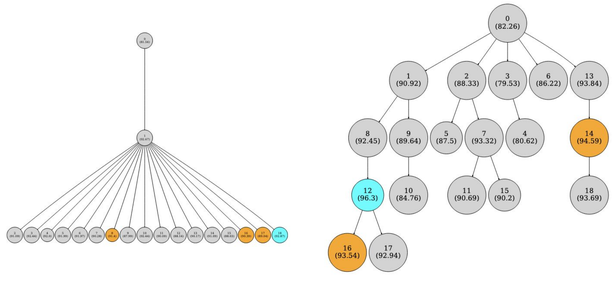

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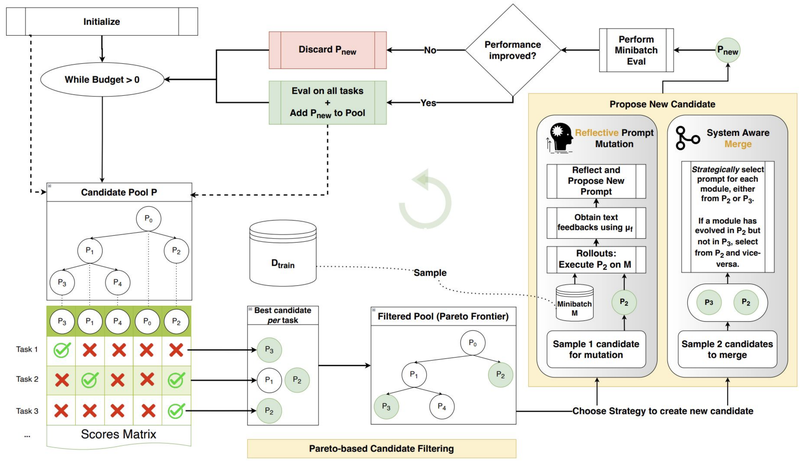


Examples

Reward Design with Language Models

Minae Kwon, Sang Michael Xie, Kalesha Bullard, Dorsa Sadigh

Goal

Make it *easier* to communicate our objectives to RL agents by using natural language

I want a versatile negotiator

Description

Alice : propose: book=1 hat=1 ball=0
Bob : propose: book=0 hat=1 ball=0
Alice : propose: book=1 hat=0 ball=1
Agreement!
Alice : 4 points
Bob : 5 points

Alice is versatile because she suggested different proposals

Handful of Examples w. Explanation

Prompt

Task Description

Alice and Bob are negotiating how to split a set of books, hats, and balls.

Example from User Describing Objective

Alice : propose: book=1 hat=1 ball=0
Bob : propose: book=0 hat=1 ball=0
Alice : propose: book=1 hat=0 ball=1

Is Alice a versatile negotiator?
Yes, because she suggested different proposals.

Episode Outcome

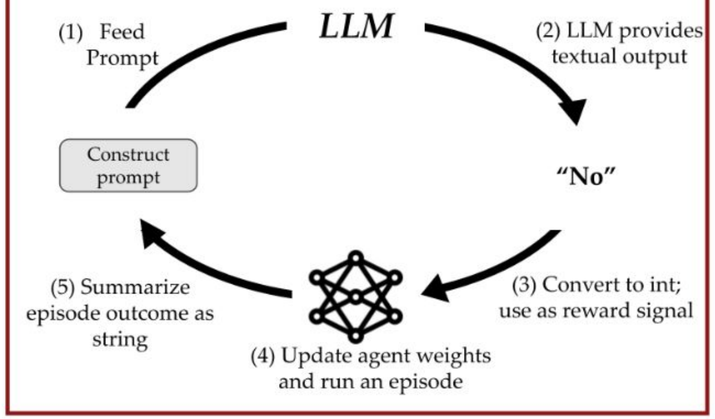
Alice : propose: book=1 hat=1 ball=0
Bob : propose: book=0 hat=1 ball=0
Alice : propose: book=1 hat=1 ball=0

Question

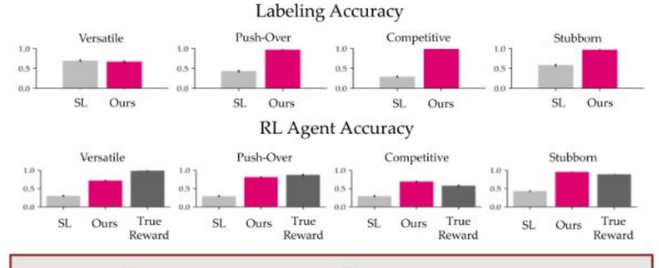
Is Alice a versatile negotiator?

Our Framework

Key Insight: Use LLM as a proxy reward



Automatic Evaluation

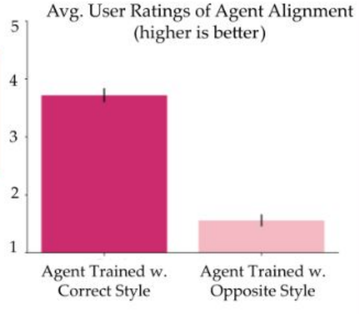


Takeaway: ✅Outperform SL by 46% ✅Underperform True Reward by 4%

Pilot User Evaluation

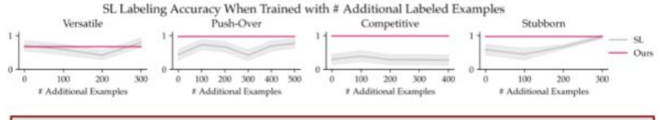
Takeaway:
✅ Humans find our agents to be more aligned than an agent trained with a different objective.

Examples of user chosen styles: ambitious, polite, considerate



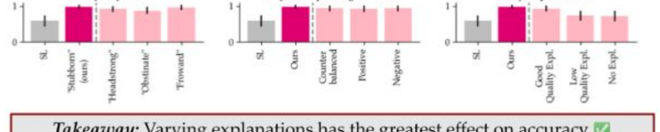
Further Analysis

How much data is required?



Takeaway: SL needs on the order of 100s of more examples to be accurate ✅

How much prompt design is required?



Takeaway: Varying explanations has the greatest effect on accuracy ✅